

- 10 C** mass on moon needs just enough KE to reach location of 0 field strength, then it will accelerate towards earth "from rest":

loss in KE = gain in EPE

$$\frac{1}{2} \cancel{m} v^2 - 0 = \cancel{m} (\Delta\phi)$$

$$v^2 = 2(\text{smaller area})$$

$$v = \sqrt{2(2.6 \times 10^6)}$$
$$= 2280 \text{ m s}^{-1}$$

- 11 B** satellite is now further away from earth so GPE must increase